Department of Physics

PCS 125 – Waves and Fields

Please insert in the circle the **first letter** of your **last name**

Please Print Clearly:

Student: Solutions _____
Family Name *First Name*

Student Number: _____ **Student Section:** _____

Final EXAM – April 14, 2014

Put a check mark in the box in front of the name of your professor:

Dr. Carvalho		Dr. Hu		Dr. Manzhos		Dr. Xu		Dr. Yuan	

Read the following attentively

- **Duration of the test: 3 hours.**
- **The test consists of 26 multiple-choice questions (1 mark each). Answer ALL questions.**
- **For each multiple choice question:**
  - **You need to show, in the space provided in this booklet, a justification for your answer. If no work or correct argument is shown, the resultant mark may be zero, even if the answer in the bubble sheet is correct.**
  - **The work shown in your exam booklet may be checked, <u>at least for one</u> of the questions, selected randomly.**
  - **<u>Transfer your choices to the bubble sheet only when you're sure of them.</u>**
  - **Some answers may be rounded off.**
- **Pagers and cell phones must be silenced and placed out of reach. Hats and earphones are not allowed. If a phone rings, its owner is responsible for disrupting the exam and the phone will be confiscated or the owner will be asked to leave the room resulting in a zero on the test.**
- **Sharing of pens, erasers and calculators is not permitted.**
- **Talking to another student or glancing over another student's paper is not permitted and may result in a charge of academic misconduct.**
- **Formula sheet provided with the test booklet. No other sheets allowed.**
- **The only calculators allowed are either the Sharp EL-546 or Casio FX-991. If you are found to have another type of calculator (even if not programmable) <u>it will be confiscated</u> for the duration of the test.**

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Student signature (no signature, no mark!)

Question 1.

A body oscillates in simple harmonic motion along the x axis. Its displacement (in m) varies with time according to the equation $x = 5.0 \sin(\pi t + \pi/3)$. The velocity (in m/s) of the body at $t = 1.0$ s is

a. +7.9
b. −7.9
c. −14
d. +14
e. −5.0

$$x = 5.0 \sin\left(\pi t + \frac{\pi}{3}\right)$$

$$v(t) = \frac{dx}{dt} = 5.0\pi \cos\left(\pi t + \frac{\pi}{3}\right)$$

$$v(1) = 5.0\pi \cos\left(\pi \cdot 1 + \frac{\pi}{3}\right) = 5.0\pi \cos\frac{4\pi}{3} =$$

$$= 5.0\pi\left(-\frac{1}{2}\right) = -2.5\pi = -7.9$$

$$x = A\cos\left(\sqrt{\frac{k}{m}}\,t\right) = 0.07\text{ m}\cos\left(\sqrt{98 s^{-2}}\,t\right)$$

$$\approx 0.07\text{m} \cdot \cos 9.9t = 0.07 \cos 9.9t$$

Question 2.

A body of mass 5.0 kg is suspended by a spring which stretches 10 cm when the mass is attached to it. It is then displaced downward an additional 7.0 cm and released at t=0 s. Its position (in m) as a function of time is

a. $y = 0.07 \sin(9.9t)$
b. $y = 0.17 \cos(9.9t)$
c. $y = 0.10 \cos(9.9t + 0.1)$
d. $y = 0.10 \sin(9.9t + 5.0)$
e. $y = 0.07 \cos(9.9t)$

When a body of mass m is suspended from a spring the spring is stretched by distance Δl.

In equilibrium $mg = k\Delta l \Rightarrow \frac{k}{m} = \frac{g}{\Delta l} = \frac{9.8\frac{m}{s^2}}{0.1m} = 98 s^{-2}$.

When a body of mass m is displayed by an additional distance x_0 and released from rest at $t=0$, the body is set up in harmonic oscillations $x = A\cos(\omega t + \varphi)$ with amplitude equal to the distance x_0 ($A = x_0$), frequency $\omega = \sqrt{\frac{k}{m}}$, and initial phase $\varphi = 0$.

Question 3.

8 per 60 seconds

Ocean waves with a wavelength of 120 m are coming in at a rate of 8 per minute. What is the wave speed?

a. 8.0 m/s
b. 16 m/s
c. 24 m/s
d. 30 m/s
e. 4.0 m/s

$\lambda = 120\,m$

$f = \frac{8}{60s} = 0.13\,s^{-1}$

$v = \lambda f = 120\,m\,(0.13\,s^{-1}) = 15.6\,\frac{m}{s} \simeq 16\,\frac{m}{s}$

Question 4.

For the wave described by $y(x,t) = 0.15\,\sin[\frac{\pi}{16}(2x - 64t)]$ (in SI units), determine the first positive x coordinate where y is a positive maximum when $t = 0$.

a. 16 m
b. 8 m
c. 4 m
d. 2 m
e. 13 m

$y(x,t) = 0.15\,\sin\left[\frac{\pi}{16}(2x - 64t)\right]$

$y(x,0) = 0.15\,\sin\frac{\pi}{16}\cdot 2x = 0.15\,\sin\frac{\pi}{8}x$.

The first positive maximum is reached at x found from equation $\frac{\pi}{8}x = \frac{\pi}{2} \Rightarrow 4\,m$.

$$I = \frac{P_{avg}}{A} = \frac{P}{4\pi r^2}.$$

Question 5.

A howler monkey is the loudest land animal and, under some circumstances, can be heard up to a distance of 5.0 km. Assume the acoustic output of a howler to be uniform in all directions and that the threshold of hearing is 1.0×10^{-12} W/m^2. The acoustic power emitted by the howler is closest to

a. 0.31 mW
b. 0.11 mW
c. 1.1 mW
d. 3.2 mW
e. 11 mW

$$P = I_0 A = I_0 (4\pi r^2) =$$

$$= 1.00 \times 10^{-12} \frac{W}{m^2} \left(4\pi \times (5.0 \times 10^3 \text{ m})^2\right) =$$

$$= 4(25) \cdot \pi \cdot 10^6 \times 10^{-12} \text{ W} = \pi \times 10^{-4} \text{ W} =$$

$$= 3.14 \times 10^{-4} \text{ W} = 0.314 \times 10^{-3} \text{ W} = 0.31 \text{ mW}$$

Question 6.

A truck moving at 36 m/s passes a police car moving at 45 m/s in the opposite direction. If the frequency of the siren is 750 Hz relative to the police car, what is the frequency heard by an observer in the truck after the police car passes the truck? (The speed of sound in air is 343 m/s.)

a. 361 Hz
b. 636 Hz
c. 393 Hz
d. 396 Hz
e. 593 Hz

→ truck police car ←

police car ($V_s = 45 \frac{m}{s}$) ← → truck ($V_0 = 36 \frac{m}{s}$)

source is moving from the observer and observer is moving from the source.

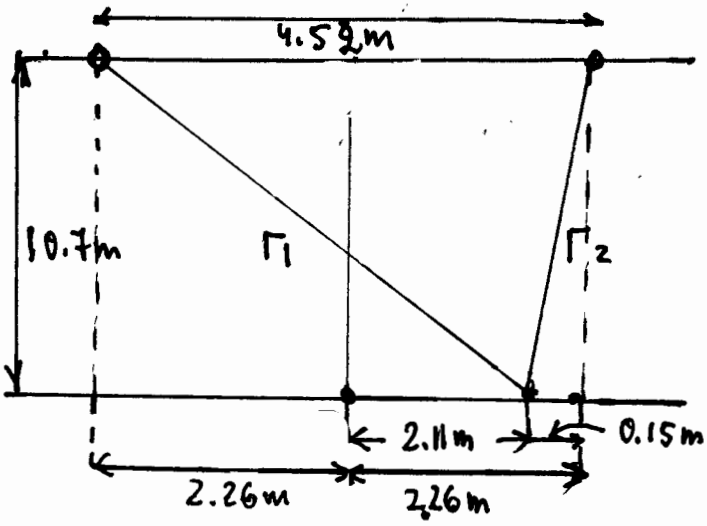
$$f = f_0 \frac{V - V_0}{V + V_s} = 750 \text{ Hz} \frac{343 \frac{m}{s} - 36 \frac{m}{s}}{343 \frac{m}{s} + 45 \frac{m}{s}} =$$

$$= 750 \text{ Hz} \frac{307}{388} \simeq 593 \text{ Hz}$$

Question 7.

Two stereo speakers mounted 4.52 m apart on a wall emit identical sound waves. You are standing at the opposite wall of the room at a point directly in front of the midpoint between the two speakers. You walk 2.11 m parallel to the wall, to a location where you first notice that the sound intensity is almost zero. If the distance between the wall along which you are walking and the wall with the speakers is 10.7 m, what is the wavelength of the sound waves?

a. 2.57 m b. 2.91 m c. 0.86 m d. 2.05 m e. 1.71 m



For destructive interference.

$\Delta r = \frac{n\lambda}{2}$, where n is odd ($n = 1, 3, \ldots$)

$n = 1 \Rightarrow \lambda = 2\Delta r$.

$r_1 - r_2 = \sqrt{(10.7m)^2 + (2.26m + 2.11m)^2} - \sqrt{(10.7m)^2 + (0.15m)^2} = 0.86m.$

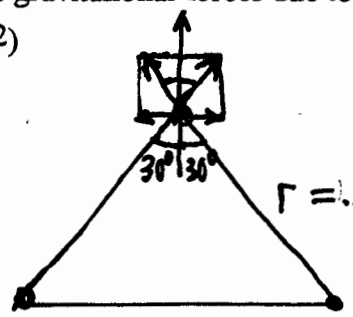
$\lambda = 2\Delta r = 1.71m$

Question 8.

Three identical very small 50 kg masses are held at the corners of an equilateral triangle, 0.30 m on each side. If one of the masses is released, what is its initial acceleration if the only forces acting on it are the gravitational forces due to the other two masses?

($G = 6.67 \times 10^{-11}$ N · m2/kg2)

a. 3.7×10^{-8} m/s^2
b. 2.5×10^{-8} m/s^2
c. 1.9×10^{-8} m/s^2
d. 4.2×10^{-8} m/s^2
e. 6.4×10^{-8} m/s^2



$r = 0.30m$

Horizontal components of the forces acting on one of the masses from two others are canceled. due to the symmetry.

Magnitude of vertical component is equal to the magnitude of the force $F_g = 2G\frac{m^2}{r^2}\cos 30^\circ \Rightarrow a = \frac{F_g}{m} = 2G\frac{m}{r^2}\cos 30^\circ =$

$= 2G\frac{m}{r^2}\frac{\sqrt{3}}{2} = 6.67 \times 10^{-11}\frac{N \cdot m^2}{kg^2} \times \frac{50\,kg}{(0.3m)^2}\sqrt{3} = 6.4 \times 10^{-8}\frac{m}{s^2}.$

a. $\frac{2}{r^2} > \frac{1}{r^2} > \frac{1}{(r+2)^2} \Rightarrow$ eq. a. does not have the solutions

b. $\frac{2}{r^2} + \frac{1}{(2-r)^2} > 0 \Rightarrow$ eq. b. does not have the solutions

c. $\frac{2}{(r+2)^2} = \frac{1}{r^2} \Rightarrow \sqrt{2}\,r = r+2 \Rightarrow (\sqrt{2}-1)r = 2 \Rightarrow \boxed{r = \frac{2}{\sqrt{2}-1}} = 5.83\,m.$

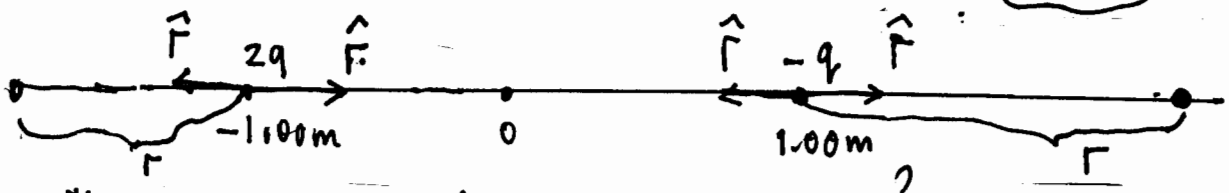
q_1 q_2

$\boxed{E = k_e \frac{q}{r^2}\hat{r}}$

Question 9.

A positive charge $+2q$ lies at $x = -1.00$ m, and a negative charge $-q$ lies at $x = 1.00$ m. The point on the x-axis where the electric field is zero is

a. 4. 83 m b. 0.18 m c. 3.00 m d. –0.33 m e. 5.83 m



The point, where electric field is 0 can be ⓐ to the left of charge 2q, ⓑ between charges and ⓒ to the right of charge -q.

ⓐ $\vec{E} = k_e \frac{2q}{r^2}(-\hat{i}) + k_e \frac{(-q)}{(r+2)^2}(-\hat{i}) = -k_e q\,\hat{i}\left(\frac{2}{r^2} - \frac{1}{(r+2)^2}\right) = 0$

b. $\vec{E} = k_e \frac{2q}{r^2}(\hat{i}) + k_e \frac{-q}{(2-r)^2}(-\hat{i}) = k_e q\,\hat{i}\left(\frac{2}{r^2} + \frac{1}{(2-r)^2}\right) = 0$

c. $\vec{E} = k_e \frac{2q}{(r+2)^2}\hat{i} + k_e \frac{(-q)}{r^2}\hat{i} = k_e \frac{q}{r}\left(\frac{2}{(r+2)^2} - \frac{1}{r^2}\right) = 0$

Question 10.

A particle ($q = +3.0$ mC, $m = 2.0$ g) has a speed of 20 m/s when it enters a region where the electric field has a constant magnitude of 80 N/C and a direction which is the same as the velocity of the particle. What is the speed of the particle 0.3 s after it enters this region?

a. 68 m/s b. 44 m/s c. 56 m/s d. 80 m/s e. 36 m/s

$$V = V_0 + a\Delta t = V_0 + \frac{qE}{m}\Delta t =$$

$$= 20\frac{m}{s} + \frac{3.0 \times 10^{-3}\,C \times 80\frac{N}{C}\,(0.3\,s)}{2 \times 10^{-3}\,kg} =$$

$$= 20\frac{m}{s} + 36\frac{m}{s} = 56\frac{m}{s}$$

Question 11.

A proton (mass $m = 1.67 \times 10^{-27}$ kg, charge $q = 1.60 \times 10^{-19}$ C) moves from point A to point B under the influence of an electric force only. At point A the proton moves with a speed of 6.0×10^4 m/s and at point B the speed of the proton is 8.0×10^4 m/s. Determine the potential difference $V_B - V_A$.

a. +15 V
b. −15 V
c. − 33 V
d. +233V
e. − 20 V

$$\Delta V = V_B - V_A = \frac{\Delta U}{q} = \frac{-\Delta K}{q} = \frac{-\frac{m}{2}(V_B^2 - V_A^2)}{q}$$

$$= -\frac{m(V_B^2 - V_A^2)}{2q} =$$

$$= -\frac{1.67\times10^{-27}\,kg\,(8.0^2 - 6.0^2)\times(10^4\,\frac{m}{s})^2}{2\,(1.6\times10^{-19}\,C)} =$$

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$$= -\frac{1.67\,(28)}{2\,(1.6)}\,\underbrace{10^{-27+8+19}}_{1}\,\frac{kg\cdot\frac{m^2}{s^2}}{C} = -14.6\,\frac{J}{C} = -14.6\,V$$

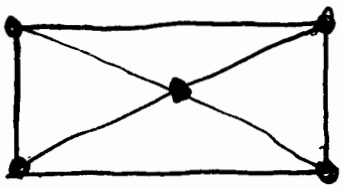
$N\cdot m = J$

$$= -15\,V.$$

Question 12.

Four identical point charges (+6.0 nC) are placed at the corners of a rectangle which measures 6.0 m × 8.0 m. If the electric potential is taken to be zero at infinity, what is the potential at the geometric center of this rectangle?

a. 58 V
b. 63 V
c. 43 V
d. 84 V
e. 11 V



$$V = 4k_e\frac{q}{r} = 4\left(8.988\times10^9\,\frac{N\cdot m^2}{C^2}\right)\frac{6.0\times10^{-9}\,C}{5.0\,m} =$$

$$= 43\,\frac{N\cdot m}{C} = 43\,\frac{J}{C} = 43\,V$$

Question 13.

Two identical particles, each with a mass of 2.0×10^{-6} kg and a charge of 25 nC, are released simultaneously from rest when the two are 4.0 cm apart.
What is the speed of either particle at the instant when the two are separated by 10 cm?

a. 7.3 m/s
b. 9.8 m/s
c. 9.2 m/s
d. 6.5 m/s
e. 4.6 m/s

$$\Delta K = -\Delta U \qquad U = k_e \frac{q^2}{r}$$

$$\frac{mv^2}{2} + \frac{mv^2}{2} = -\left(k_e \frac{q^2}{r_f} - k_e \frac{q^2}{r_i}\right) \Rightarrow$$

$$mv^2 = k_e q^2 \left(\frac{1}{r_i} - \frac{1}{r_f}\right) \Rightarrow \qquad \frac{N \cdot m^2}{kg \cdot \frac{1}{m}} = \frac{kg \cdot \frac{m}{s^2}}{kg \cdot m^{-1}} = \frac{m^2}{s^2}$$

$$v = q\sqrt{\frac{k_e}{m}\left(\frac{1}{r_i} - \frac{1}{r_f}\right)} = 25 \times 10^{-9} C \sqrt{\frac{8.888 \times 10^9 \frac{N \cdot m^2}{C^2}}{2.0 \times 10^{-6} kg}\left(\frac{1}{4 \times 10^{-2} m} - \frac{1}{10 \times 10^{-2} m}\right)}$$

$$\frac{100}{4} - \frac{100}{10} = 15$$

$$= 25 \times 10^{-9} \times 10^8 \sqrt{\frac{8.888}{20} \cdot 15} \ \frac{m}{s} = 64.5 \times 10^{-1} \frac{m}{s} = 6.5 \frac{m}{s}$$

Question 14.

A current of 1.00 A flows in a copper wire with cross-section 4×10^{-5} m^2. with electron density 8.49×10^{28} electrons/m^3.
What is the magnitude and direction of the electron drift velocity, with the direction defined relative to the direction of the current?

a. -1.84×10^{-6} m/s
b. $+1.84 \times 10^{-6}$ m/s
c. $+5.43 \times 10^{-10}$ m/s
d. -5.43×10^{-6} m/s
e. $+5.43 \times 10^{-6}$ m/s

the velocity of electrons is opposite to the direction of the current

$$V_d = -\frac{I_{avg}}{A|q|n} = -\frac{1A}{4 \times 10^{-5} m^2 (1.6 \times 10^{-19} C)(8.49 \times 10^{28} m^{-3})}$$

$$= -\frac{10^{-28+5+19}}{4(1.6)(8.48)} \frac{m}{s} = -0.0184 \times 10^{-4} \frac{m}{s} = -1.84 \times 10^{-6} \frac{m}{s}$$

$$T = \frac{N}{\frac{m}{s}\cdot C} \Rightarrow \frac{kg}{C\cdot T} = \frac{kg}{C\cdot \frac{N}{\frac{m}{s}\cdot C}} = \frac{kg\cdot\frac{m}{s}}{kg\cdot\frac{m}{s^2}} = s$$

Question 15.

An electron follows a circular path of radius 15.0 cm in a uniform magnetic field of 0.0003T. What is the period of this motion?

a. 0.18 μs b. 1.2 ms c. 0.12 μs d. 1.8 ms e. 1.8 μs

$$T = \frac{2\pi r}{v}$$

$$\frac{mv^2}{r} = qvB \Rightarrow \frac{v}{r} = \frac{qB}{m} \text{ and } T = \frac{2\pi m}{qB} =$$

$$= \frac{2\pi(9.1\times10^{-31}kg)}{(1.6\times10^{-19}C)(3\times10^{-4}T)} = \frac{2\pi(9.1)}{1.6(3)}10^{-31+19+4} = 1$$

$$= 11.9\times10^{-8}C = 0.119\times10^{-6}s = 0.12\ \mu s$$

Question 16.

A Hall-effect probe operates with a 100 mA current. When the probe is placed in a uniform magnetic field of magnitude 0.065 T, it produces a Hall voltage of 0.75 μV. When it is used to measure an unknown magnetic field, the Hall voltage is 0.45μV. What is the magnitude of the unknown field?

a. 9.8 mT b. 39 mT c. 3.9 mT d. 98 ms e. 108 mT

$$\Delta V_{H_1} = \frac{nqt}{I}B_1 \Rightarrow \frac{nqt}{I} = \frac{\Delta V_{H1}}{B_1}.$$

$$B_2 = \Delta V_{H_2}\left(\frac{I}{nqt}\right) = \frac{\Delta V_{H2}}{\Delta V_{H1}}B_1$$

$$B_2 = \frac{0.45\mu V\ (0.065T)}{0.75\mu V} = 0.039T = 39\ mT$$

Question 17.

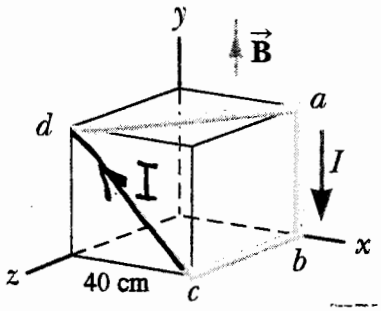
The cube shown in the figure is 40 cm on each edge. Four segments of conducting wire (*ab*, *bc*, *cd*, and *da*) form a closed loop that carries a current $I = 5A$ in the direction shown. A magnetic field of magnitude $B = 20$ mT exists in this region and its direction is in the positive y direction. Find the magnetic force vector on the wire segment *cd*.

a. 40**i** mN
b. –40**i** mN
c. 40 **k** mN
d. –40 **k** mN
e. (40**i** + 40**k**) mN

$\vec{F} = I\vec{L} \times \vec{B}$

$\vec{L} = -0.40\hat{i}\,m + 0.40\hat{j}\,m$

$\vec{B} = 20\,mT\,\hat{j}$



$$\vec{F} = I\vec{L} \times \vec{B} = 5A \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -0.40 & 0.40 & 0 \\ 0 & 20\times10^{-3} & 0 \end{vmatrix} =$$

$$= 5A\,(-0.40 \times 20 \times 10^{-3})\,\hat{k}\ T\cdot m = -40\times10^{-3}\,\hat{k}\ \cancel{A}\cdot\frac{N}{\cancel{m}/s}\cdot\cancel{C}\cdot\cancel{m} =$$

$$= -40\times10^{-3}\,\hat{k}\ N = -40\,\hat{k}\ mN$$

Question 18.

A particle (mass = 2.0 mg, charge = –6.0 μC) moves in the positive direction along the x axis with a velocity of 3.0 km/s. It enters a magnetic field of (2**i** + 3**j** + 4**k**) mT. What is the acceleration of the particle (in m/s^2)?

a. (–36 **j** +27 **k**) b. (36 **j** –27 **k**) c. (–24 **j** +18 **k**) d. (24 **j** –18 **k**)
e. None of these (give value)

$$\vec{a} = \frac{\vec{F}}{m} = \frac{q}{m}\left[\vec{v}\times\vec{B}\right] = \frac{-6.0\times10^{-6}\,C}{2.0\times10^{-6}\,kg}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3\times10^{3} & 0 & 0 \\ 2\times10^{-3} & 3\times10^{-3} & 4\times10^{-3} \end{vmatrix}\frac{m}{s}T$$

$$= -3\left(-12\hat{j} + 9\hat{k}\right)\frac{\cancel{C}}{\cancel{kg}}\cdot\frac{\cancel{m}}{\cancel{s}}\cdot\frac{N}{\cancel{C}\cdot\frac{\cancel{m}}{\cancel{s}}} = \left(36\hat{j} - 27\hat{k}\right)\frac{m}{s^2}$$

Question 19.

Two long parallel wires are separated by 4.0 cm. One of the wires carries a current of 20 A and the other carries a 30-A current. Determine the magnitude of the magnetic force on a 2.0-m segment of the wire carrying the greater current.

a. 7.0 mN
b. 6.0 mN
c. 8.0 mN
d. 9.0 mN
e. 4.0 mN

$$F = \mu_0 \frac{I_1}{2\pi a} \cdot I_2 \ell = \mu_0 \frac{I_1 I_2}{2\pi a} \ell =$$

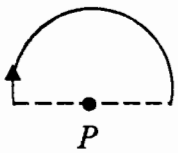
$$= \cancel{4\pi}^{2} \times 10^{-7} \frac{T \cdot m}{\cancel{A}} \cdot \frac{30\cancel{A}(20A)}{\cancel{2\pi}(0.04\cancel{m})}(2\cancel{m}) =$$

$$= 60{,}000 \times 10^{-7} \frac{N}{\frac{m}{s} \cdot C} \cdot m \cdot \frac{C}{s} = 6.0 \times 10^{-3} N = 6.0 \text{ mN}$$

Question 20.

A segment of wire of total length 3.0 m is formed into a semi-circle as shown in the figure. The wire carries a 15 A current in the direction indicated by the arrow. Determine the magnitude of the magnetic field at the center if the circle point *P*.

a. 1.0 μT
b. 2.9 μT
c. 5.0 μT
d. 9.8 μT
e. None of the above (give the value)



$\ell = 3.0 m.$

$$B = \frac{\mu_0 I \theta}{4\pi a}, \quad \theta = \pi \Rightarrow B = \frac{\mu_0 I}{4a}.$$

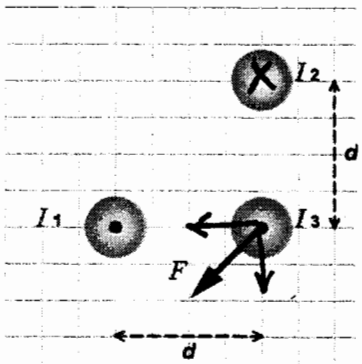
$$a = \frac{\ell}{\pi} \Rightarrow B = \frac{\mu_0 I \pi}{4\ell} = \frac{\cancel{4}\pi \times 10^{-7} \frac{T \cdot \cancel{m}}{\cancel{A}} (\cancel{15}^{5}\cancel{A})\pi}{\cancel{4}(\cancel{3.0}\cancel{m})} =$$

$$= 5.0\pi^2 \times 10^{-7} T \simeq 49 \times 10^{-7} T = 4.9 \times 10^{-6} T =$$

$$= 4.9 \mu T \simeq 5.0 \mu T$$

Question 21.

The figure shows three long, parallel, current-carrying wires. The current directions are indicated for currents I_1 and I_3. The arrow labeled F represents the net magnetic force acting on current I_3. The three currents have equal magnitudes. What is the direction of the current I_2?



a. into the picture (in the direction opposite to that of I_1 and I_3)
b. horizontal to the right
c. vertically upward
d. vertically downward
e. out of the picture (in the same direction as I_1 and I_3)

The direction of the current in the wire with currents I_3 and I_1 are the same ⇒ the wire with current I_3 is attracted to I_1. This means that the force acting from I_2 is directed downward and therefore the direction of the current I_2 is opposite to the direction of I_1 and I_3

Question 22.

A standing wave on a string fixed at both ends vibrates in its second harmonic. Suppose the tension is changed to four times the previous value while the frequency and the length of the string are held constant. Which standing wave pattern is produced?

a. no standing wave for this condition
b. first harmonic
c. second harmonic
d. fourth harmonic
e. eighth harmonic

$$f_n = \frac{v}{\lambda_n} = \frac{n\sqrt{\frac{T}{\mu}}}{2L} \cdot n = 1, 2, 3, \ldots$$

$n = 2.$

$$f_2 = \frac{1}{L}\sqrt{\frac{T}{\mu}}.$$

$$\lambda_n = \frac{v}{f} = \frac{\sqrt{\frac{4T}{\mu}}}{\frac{1}{L}\sqrt{\frac{T}{\mu}}} = \frac{2}{\frac{1}{L}} = 2L \Rightarrow n = 1$$

Question 23.

Consider the following three statements pertaining to static electric field lines

(i) Electric field lines can be either straight or curved. ✓
(ii) Electric field lines begin on positive charges and end on negative charges. ✓
(iii) Electric field lines can intersect with one another. ✗

a. Only (i) is true ✗
b. Only (ii) is true ✗
c. Only (iii) is true ✗
d. Only (i) and (ii) are true ✓
e. Only (ii) and (iii) are true ✗

For the planets on the orbits. $\frac{mv^2}{r} = \frac{GMm}{r^2} \rightarrow K = \frac{mv^2}{2} =$

$= \frac{GMm}{2r} = -\frac{1}{2}\left(-\frac{GMm}{r}\right) = -\frac{1}{2}U > 0 \Big| E = -\frac{1}{2}U + U = \frac{1}{2}U < 0$

$V = -\frac{GmM}{r}$ $K = \frac{GmM}{2r}$ $E = -\frac{1}{2}\frac{GMM}{2r}$ $r_A < r_B$.

Question 24.

Two planets having equal masses are in circular orbit around a star. Planet *A* has a smaller orbital radius than planet *B*. Which statement is true?

V

a. Planet *A* has more kinetic energy, less potential energy, and less total energy (potential plus kinetic) than planet *B*.

X b. Planet *A* has more kinetic energy, more potential energy, and more total energy (potential plus kinetic) than planet *B*.

X c. Planet *A* has more kinetic energy, less potential energy, and more total energy (potential plus kinetic) than planet *B*.

X d. Planet *A* and planet *B* have the same amount of total energy (potential plus kinetic).

X e. None of the above (Explain your answer)

Question 25.

When a positive charge is released from rest and moves along an electric field line, it moves to a position of

$q > 0$

$V = \frac{U}{q}$.

K increases and U decreases

V also decreases.

X a. lower electric potential and higher electric potential energy

V b. lower electric potential and lower electric potential energy

X c. higher electric potential and lower electric potential energy

X d. higher electric potential and higher electric potential energy

e. None of the above

Question 26.

Equal charges, one at rest, the other having a velocity of 10^4 m/s, are released in a uniform magnetic field. Which of the following charges has the largest force exerted on it by the magnetic field?

X a. The charge that is at rest.

X b. The charge that is moving, if its velocity is parallel to the magnetic field direction when it is released.

X c. The charge that is moving if its velocity makes an angle of 45° with the direction of the magnetic field when it is released.

d. The charge that is moving if its velocity is perpendicular to the magnetic field direction when it is released.

X e. All the charges above experience equal forces when released in the same magnetic field.

PCS125 Constants and Formulae

$I_0 = 1.00 \times 10^{-12}\text{W/m}^2$ $\quad$ $c = 3 \times 10^8$ m/s

$\rho_{\text{air}} = 1.2$ kg/m^3 $\quad$ $G = 6.6742 \times 10^{-11}\text{N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$

$g = 9.8 \ \text{m} \cdot \text{s}^{-2}$ $\quad$ $k = 8.99 \times 10^9$ N $\cdot$m$^2\cdot$ C^{-2}

$e = 1.6 \times 10^{-19}$ C $\quad$ 1 eV$=1.6\times10^{-19}$ J

$m_e = 9.1094 \times 10^{-31}$ kg $\quad$ $m_p = 1.6726 \times 10^{-27}$ kg

$\mu_0 = 4\pi \times 10^{-7}$ T$\cdot$m$\cdot$ A^{-1}

$\sin a \pm \sin b = 2\cos\left(\frac{a \mp b}{2}\right)\sin\left(\frac{a \pm b}{2}\right)$ $\quad$ $\sin(a \pm b) = \sin a \cos b \pm \cos a \sin b$

$\omega_0 = \sqrt{\frac{k}{m}}$ $\quad$ $\omega_0 = \sqrt{\frac{g}{L}}$ $\quad$ $x = A\exp\left(-\frac{b}{2m}t\right)\cos(\omega t)$ $\quad$ $\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$

$A = \frac{F_0/m}{\sqrt{\left(w^2-\omega_0^2\right)^2+(b\omega/m)^2}}$ $\quad$ $v = 331\sqrt{1+\frac{T_c}{273}}$ m/s $\quad$ $\beta = 10\log\frac{I}{I_0}$

$B = \frac{-\Delta P}{\Delta V/V_i}$ $\quad$ $v = \sqrt{\frac{T}{\mu}}$ $\quad$ $v = \sqrt{Y/\rho}$ $\quad$ $v = \sqrt{B/\rho}$

$s(x,t) = s_{\text{max}}\cos(kx - \omega t)$ $\quad$ $\Delta P_{\text{max}} = \rho v \omega s_{\text{max}}$ $\quad$ $\wp = \frac{1}{2}\rho A\omega^2 s_{\text{max}}^2 v$

$I = \frac{\wp}{A} = \frac{\wp}{4\pi r^2}$ $\quad$ $I = \frac{\Delta P_{\text{max}}^2}{2\rho v}$ $\quad$ $\wp = \frac{1}{2}\mu\omega^2 A^2 v$ $\quad$ $f' = \left(\frac{v+v_O}{v-v_S}\right)f$

$f_B = |f_1 - f_2|$ $\quad$ $F = G\frac{m_1 m_2}{r^2}$ $\quad$ $F = k\frac{q_1 q_2}{r^2}$ $\quad$ $F = q\vec{v} \times \vec{B}$

$W_{A\to B}^{\text{field}} = \int_A^B \vec{F}\cdot d\vec{r}$ $\quad$ $U_A - U_B = q(V_A - V_B) = W_{A\to B}^{\text{field}}$ $\quad$ $U(r) = -\frac{GMm}{r}$

$I = nqAv_D$ $\quad$ $V_H = \frac{IB}{nqt}$

$$\vec{B} = \frac{\mu_0 I}{4\pi}\int\frac{d\vec{s}\times\hat{r}}{r^2}$$

$B = \frac{\mu_0 I}{2\pi a}$ (at a distance a from an infinitely long wire)

$B = \frac{\mu_0 I\theta}{4\pi a}$ (at the centre of an arc wire of radius a subtended by an angle θ)

$B = \frac{\mu_0 I a^2}{2\left(a^2+x^2\right)^{3/2}}$ (at a distance x along the centre line axis of a circular loop of wire of radius a)